

Waypoint

AI Trip Planning Assistant

Framework: Phidata + Streamlit | Version: 1.0 | Date: June 2026
Repository: bibhu2020/agenticaai (Hugging Face Space)

1. Overview

Waypoint is a comprehensive AI trip planner that gathers real-time data before generating an itinerary. It uses the Phidata agent framework with nine specialised tools covering weather, attractions, restaurants, activities, transportation, expense calculation, and currency conversion. Google Places is the primary data source with Tavily Search as an automatic fallback.

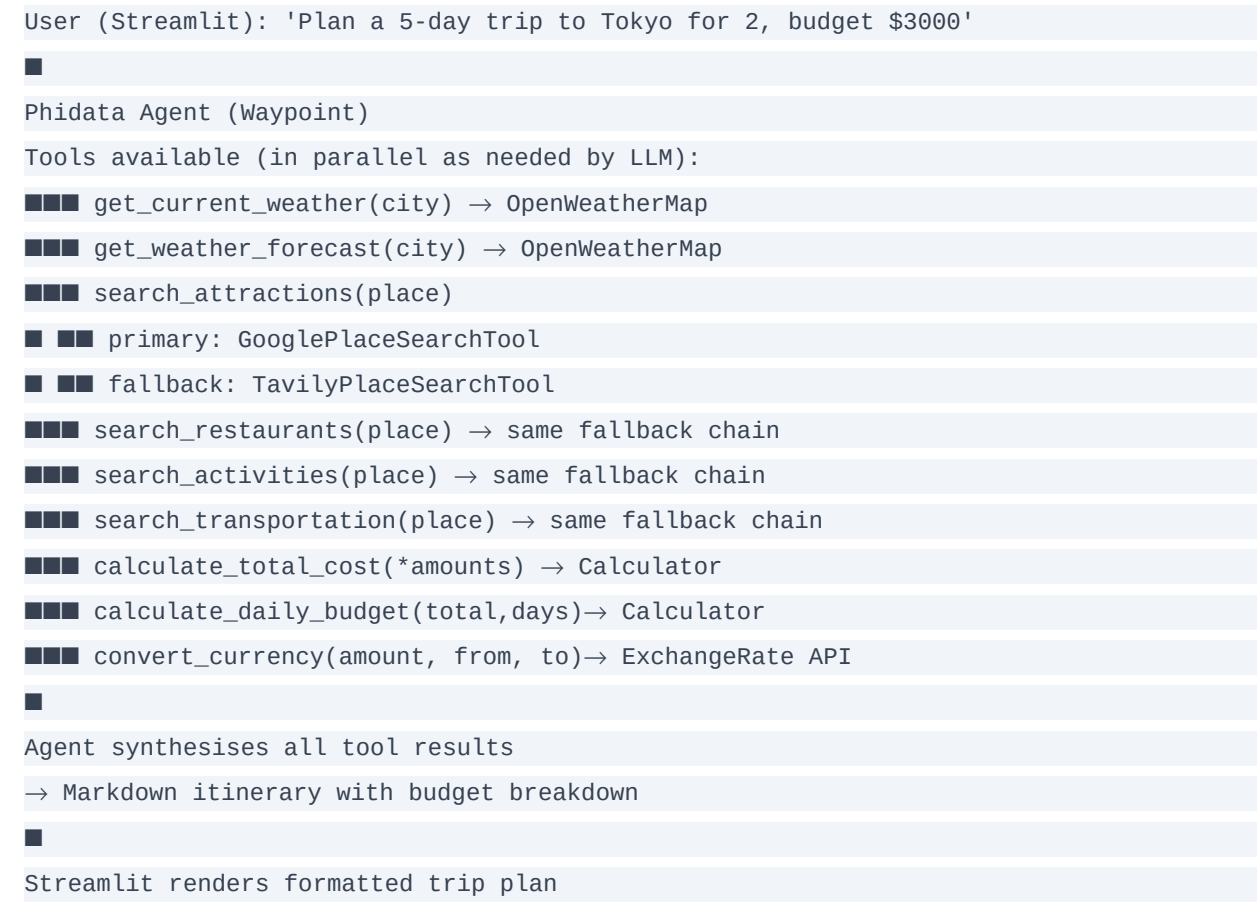
2. System Design

2.1 Components

Component	Technology	Purpose
main.py	Streamlit	Chat UI — collects destination, budget, dates; streams response
agent/agentic_workflow.py	Phidata Agent	Single orchestrating agent with all 9 tools
utils/weather_info.py	OpenWeatherMap API	Current weather and multi-day forecast
utils/place_info_search.py	Google Places API / Tavily	Attractions, restaurants, activities, transport search
utils/expense_calculator.py	Pure Python	Total cost and daily budget calculations
utils/currency_converter.py	ExchangeRate API	Live currency conversion

Component	Technology	Purpose
llm.py	Phidata model wrapper	LLM factory (openai OpenAIChat / google Gemini)

2.2 Architecture Diagram



3. Agentic Flow

Waypoint uses a single-agent ReAct-style loop. The LLM autonomously decides which tools to call and in which order based on the user's query. Typical execution for a trip planning request:

LLM Turn	Tool Called	Purpose
1	get_weather_forecast	Understand climate for trip dates
2	search_attractions	Find top things to do
3	search_restaurants	Find dining options
4	search_activities	Find paid/free activities with pricing
5	search_transportation	Local transport costs
6	calculate_total_cost	Add up accommodation + activities + food + transport
7	calculate_daily_budget	Divide by trip duration

LLM Turn	Tool Called	Purpose
8	convert_currency	Convert to user's home currency if requested
Final	(no tool)	Generate full Markdown itinerary with all gathered data

4. Fallback Strategy

- Google Places API → primary for all place searches (attractions, restaurants, activities, transport)
- Tavily Search → automatic fallback when Google Places call raises an exception
- All fallbacks are transparent to the LLM and the user

5. Configuration

Env Var	Required for
OPENAI_API_KEY	Default LLM (gpt-4o-mini via Phidata OpenAIChat)
GOOGLE_API_KEY	Gemini model option
GPLACES_API_KEY	Google Places search (primary)
OPENWEATHERMAP_API_KEY	Weather tools
EXCHANGE_RATE_API_KEY	Currency conversion